MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF ASSISTANT DIRECTOR (TOWN PLANNING) UNDER URBAN DEVELOPMENT & POVERTY ALLEVIATION DEPARTMENT, GOVERNMENT OF MIZORAM.

(FEBRUARY-2026)

TECHNICAL SUBJECT PAPER-III

Time Allowed : 2 hours FM : 100

All questions carry equal mark of 1 each.
Attempt all questions.

1. Stormwater drains are designed primarily to:

(a) Supply irrigation water (b) Remove surface runoff
(c) Recharge reservoirs (d) Accommodate heavy rainfall

2. The full form of 'LPCD' in water supply planning is:

(a) Liters Per Capital Demand (b) Liters Per Capita per Day
(c) Liquid Pressure Central Distribution (d) Local Population Consumption per Day

3. The 'Peak Factor' in water demand calculation is used to:

(a) Reduce the pipe diameter (b) Account for hourly variations
(c) Calculate day maximum flow (d) Measure water purity

4. 'Sullage' specifically excludes waste from:

(a) Kitchen sinks (b) Wash basins
(c) Water closets (d) Bathing areas

5. 'Leachate' is a hazardous liquid primarily associated with:

(a) Water treatment filters (b) Solid waste landfills
(c) Power transformers (d) Sewerage Treatment Plan

6. According to URDPFI, a 'Primary Health Centre' (PHC) serves a population of:

(a) 5,000 (b) 30,000
(c) 50,000 (d) 70,000

7. The Self-cleansing velocity in sewers prevents:

(a) Deposition of solids (b) Bursting of pipes
(c) Excessive pipe corrosion (d) Entry of rainwater

8. A Dead-end System in water distribution is also known as:

(a) Gridiron system
(b) Ring system
(c) Tree system
(d) Radial system

9. Zonal Reservoirs are essential in hilly terrain to:

(a) Maintain uniform pressure across varying elevations
(b) Reduce the total water treatment cost
(c) Centralize the distribution system
(d) Store industrial chemical waste

10. Load Factor in power planning is the ratio of:

(a) Peak load to total capacity
(b) Average load to peak load
(c) Active power to reactive power
(d) Residential load to industrial load

11. Solid waste collection efficiency is most influenced by:

(a) Waste composition
(b) Color coded vehicles
(c) Disposal method
(d) Route planning

12. Water supply system design capacity is normally based on:

(a) Average annual rainfall
(b) Minimum daily demand
(c) Maximum daily demand
(d) Projected peak demand

13. Stormwater retention basins are primarily used to:

(a) Recharge groundwater
(b) Improve water quality
(c) Reduce peak runoff
(d) Prevent evaporation

14. Collection efficiency in solid waste management is reduced mainly by:

(a) Waste composition
(b) Climatic conditions
(c) Vehicle type
(d) Irregular service coverage

15. Network looping in water supply improves:

(a) Treatment quality
(b) Chemical stability
(c) Pressure reliability
(d) Meter accuracy

16. Manning's Roughness Coefficient (n) for a concrete pipe is typically:

(a) 0.001
(b) 0.013
(c) 0.045
(d) 0.110

17. Biochemical Oxygen Demand (BOD) testing typically occurs over 5 days at:

(a) 0°C
(b) 20°C
(c) 37°C
(d) 50°C

18. Air Valves in a water main should be placed at:

(a) Every low point (valley) of the pipeline
(b) The midpoint of every straight section
(c) Only at the treatment plant
(d) Every high point (summit) of the pipeline

19. Sewer exfiltration primarily affects:

(a) Pipe pressure
(b) Flow velocity
(c) Groundwater quality
(d) Treatment efficiency

20. Which pairing correctly represents the formula used for estimating peak runoff and the formula used for sizing open drains?

(a) Peak runoff – Manning's equation; Channel flow – Rational formula
(b) Peak runoff – Darcy's law; Channel flow – Chezy's equation
(c) Peak runoff – Unit hydrograph; Channel flow – Hazen–Williams formula
(d) Peak runoff – Rational formula; Channel flow – Manning's equation

21. A city zone generates 24 tonnes/day of solid waste. Collection vehicles have a capacity of 4 tonnes per trip and each vehicle can make 3 trips per day. Assuming 100% operational efficiency, how many vehicles are required?

(a) 1
(b) 2
(c) 3
(d) 4

22. A residential zone has a population of 12,000 and per-capita demand of 135 LPCD. If the maximum day factor = 1.8, what is the maximum daily demand?

(a) 2.16 MLD
(b) 2.43 MLD
(c) 2.92 MLD
(d) 3.24 MLD

23. In demographic analysis, Age-Sex Composition is visually represented by:

(a) Lorenz Curve
(b) Population Pyramid
(c) Scatter Plot
(d) Ogive Curve

24. Urbanism as a sociological concept refers to:

(a) Physical growth of urban centers
(b) A specific way of life
(c) Industrial production rate
(d) Influence of urban administration on citizens

25. Workforce participation rate represents:

(a) Employed population as % of total population
(b) Working population as % of total population
(c) Working population as % of adult population
(d) Employed population as % of households

26. Head Count Ratio (HCR) is a standard measure for:

(a) Traffic flow
(b) Poverty incidence
(c) Literacy rates
(d) Housing density

27. Which demographic stage is characterized by falling birth rates and low death rates?

(a) Stage I
(b) Stage II
(c) Stage III
(d) Stage IV

28. In urban sociology, "Gemeinschaft" refers to a society based on:

(a) Personal/Community ties
(b) Formal/Contractual ties
(c) Economic profit
(d) Legal regulations

29. Stratified sampling is preferred when:

(a) Population is homogeneous
(b) Area is small
(c) Population is heterogeneous
(d) Budget is unlimited

30. Who postulated Central Place Theory?

(a) Lewis Mumford
(b) Walther Christaller
(c) Tony Garnier
(d) Ludwig Hilberseimer

31. Threshold Population in Central Place Theory is the:

(a) Minimum support for a service
(b) Maximum distance for a customer
(c) Total capacity of a market
(d) Average density of a town

32. Gini Coefficient of 1.0 represents:

(a) Perfect equality
(b) Perfect inequality
(c) Zero population
(d) Maximum density

33. PM SVANidhi scheme mainly targets which socio-economic group?

(a) Small manufacturing workers
(b) Migrant construction labourers
(c) Urban street vendors
(d) Informal transport operators

34. DAY-NULM stands for:

(a) Deendayal Antyodaya Yojana - National Urban Livelihood Mission
(b) Deendayal Awas Yojana - National Urban Livelihood Mission
(c) Deendayal Antyodaya Yojana - National Urbanization Leveraging Mission
(d) Deendayal Awas Yojana - National Urban Livability Mission

35. In project planning, a Work Breakdown Structure (WBS) is mainly used to:

(a) Divide work into manageable components
(b) Estimate project financing
(c) Allocate legal responsibilities
(d) Fix contract conditions

36. The main advantage of PERT over CPM is its ability to:

(a) Reduce project cost
(b) Handle uncertainty in activity duration
(c) Improve labor productivity
(d) Simplify contract administration

37. In infrastructure projects, land acquisition risk is usually classified as:

(a) Financial risk
(b) Technical risk
(c) Operational risk
(d) Legal risk

38. · In the Pre-Feasibility Stage, which analysis is conducted to determine if a project is worth the investment of a detailed feasibility study?

(a) Sensitivity Analysis
(b) Opportunity Study
(c) Post-Mortem Analysis
(d) Impact Mitigation Audit

39. In the LARR Act 2013, 'Solatium' is defined as:

(a) The cost of building structures
(b) Extra compensation for the 'compulsory nature' of acquisition
(c) Additional tax on the profit from land sale
(d) The fee paid to the surveyor

40. A Master Plan prepared under planning legislation is:

(a) A financial document
(b) A statutory land-use plan
(c) A development manual
(d) A transport policy note

41. The 73rd Constitutional Amendment Act mainly relates to:

(a) Urban local governance
(b) Metropolitan planning
(c) Rural local governance
(d) Regional development authorities

42. A 'Nagar Panchayat' is a specific type of local body designed for:

(a) A large urban area with millions of residents
(b) A rural village with no commercial activity
(c) An industrial township with mostly rural residents
(d) A transitional area moving from rural to urban

43. One structural weakness of the decentralization framework under the 74th Amendment is that:

(a) Municipal functions are constitutionally rigid
(b) Planning powers are automatically transferred to cities
(c) Metropolitan Planning Committees replace State planning boards
(d) Financial devolution depends largely on State discretion

44. As per the 74th Amendment, how many functional items are listed under the purview of Municipalities?

(a) 29 items
(b) 18 items
(c) 12 items
(d) 24 items

45. What is the design period for the water treatment unit?

(a) 10 years
(b) 15 years
(c) 20 years
(d) 30 years

46. The process of burning municipal solid waste under controlled high temperatures in a specific furnace is called:

(a) Recycling
(b) Incineration
(c) Vermicomposting
(d) Burning

47. Which of the following is a method of wastewater treatment?

(a) Landfill disposal
(b) Incineration
(c) Composting
(d) Biological treatment

48. What is the term for the process of using living organisms to clean up or reduce environmental pollution?

(a) Bioremediation
(b) Overconsumption
(c) Soil erosion
(d) Greenwashing

49. 'DEWATS' is :

(a) Unit for measurement of energy used for treatment of waste water
(b) A decentralized system that uses ponds with aquatic plants to treat waste water
(c) A decentralized system for treatment of water purification of pounds
(d) A mechanical process of wastewater treatment used in some parts of Delhi.

50. As per NBC, what is the accepted benchmark for water requirement in urban areas?

(a) 135 lpcd (b) 70 lpcd

(c) 125 lpcd (d) 75 lpcd

51. This gas is released from landfills and can be used to generate electricity.

(a) Sulphur dioxide (b) Methane

(c) Ammonia (d) Nitrous oxide

52. SCADA is an advanced technology used in several cities. In which of the following sector does it find its application?

(a) Centralised heating and cooling (b) Water supply and sewage management

(c) Transportation planning (d) GIS and remote sensing

53. Which of the following can be provided with tertiary treatment for recycle and reuse?

(a) Municipal Solid Waste (b) Suspended Particulate Matter

(c) Waste water (d) Biomass

54. The average consumption of water required in factories in lpcd is :

(a) 10-15 (b) 20-30

(c) 30-45 (d) 70-80

55. What are the factors affecting per capita demand?

(a) Size of city

(b) Size of city, habit of people

(c) Cost of water, quality of water, size of city

(d) Cost of water, quality of water, size of city, habit of people

56. No landfill should be constructed within _____m of a navigable river or stream.

(a) 75 (b) 80

(c) 100 (d) 120

57. The 'active' period of landfill design, typically range from ____years depending on the availability of land area.

(a) 10 (b) 10-25

(c) 25-35 (d) 35

58. As per IS 10500-2012, the acceptable limit for chloramine content in drinking water is _____mg/l

(a) 10 (b) 8

(c) 4 (d) 25

59. Watershed is an important parameter in spatial planning. It stands for:

(a) A spatial unit with common drainage (b) Areas having different drainage

(c) An area devoid of drainage (d) Areas taken up for linking rivers

60. As per IS 1742:1960, recommended spacing of manholes on straight reaches on sewer line for diameter upto 0.9m is:

(a) 45m (b) 75m

(c) 90m (d) 120m

61. Which of the following does not help in disinfecting water?

(a) Filtration
(b) Chlorine tablets
(c) Alums
(d) Boiling

62. Which of the following helps in reduction of rain water runoff and mitigating of urban flooding?

1. Tree cover
2. Permeable concrete blocks
3. Lawns
4. Paver blocks

Which of the above statements are correct?

(a) 1, 2 and 4
(b) 1, 3 and 4
(c) 1, 2, 3 and 4
(d) 1, 2 and 3

63. Which of the following is not an accepted method for Municipal Solid Waste disposal?

(a) Sanitary land fill
(b) Open dump disposal
(c) Conversion of waste to compost
(d) Recycling

64. Turbidity is a measure of relative clarity of a liquid(water), and one of its unit of measurement is:

(a) MPN
(b) NTU
(c) Moles
(d) Grams

65. As per CPHEEO guidelines, the water demand for fire fighting is estimated by which of the following equations? (Note: P - Population in 1000s)

(a) $50\sqrt{P}$
(b) $100\sqrt{P}$
(c) $1000\sqrt{P}$
(d) $2000\sqrt{P}$

66. A combined sewer is one which transports domestic sewage and:

(a) Storm water
(b) Industrial wastewater
(c) Overhead flow
(d) Industrial wastewater and storm water

67. Consider the following statements :

In solid waste management:-

1. Density separation of solid wastes can be accomplished by air classifiers.
2. Iron recovery from solid wastes can be done by magnetic separators.
3. Aluminium separation from solid wastes can be accomplished by eddy current separators.

Which of the statements given above are correct?

(a) 1 and 2
(b) 2 and 3
(c) 1 and 3
(d) 1, 2 and 3

68. Which of the following statements is true about zero waste management?

(a) Separate collection of each kind of waste
(b) Segregation of garbage at the source
(c) Community involvement
(d) All of the above

69. The important steps in solid waste management are:

(a) Waste generation and collection
(b) On-site handling, storage and processing
(c) Transport and disposal
(d) All of the above

70. Benchmarks providing Authority for solid waste management as per URDPFI:

(a) CPHEEO & Service level benchmarking (b) SPCB

(c) Municipal (d) None of the above

71. Which Indian city is known for its 'waste-to-wealth' model and being a 'zero-landfill city'?

(a) Delhi (b) Pimpri-Chinchwad

(c) Bengaluru (d) Indore

72. Which of the following is NOT a major classified solid waste in India?

(a) Hazardous Waste (b) Plastic Waste

(c) Agricultural Waste (d) Construction & Demolition Waste

73. Identify the problems of Urbanisation in India from the following:

1. Widening socio-economic Gap.
2. Slums and Squatter Settlements
3. Shortage of Urban Services
4. Emergence of Towns as Socio-Cultural Centres
5. Environmental Degradation

(a) 1, 2, 3 and 5 (b) 1, 2, 3, 4, 5

(c) 1, 3, 4 and 5 (d) 2, 3 and 5

74. Which of the following is not a beneût of the Economies of Agglomeration?

(a) High land prices (b) Lower transportation costs

(c) Stronger local market capital (d) Accumulation of knowledge and human

75. The relationship between shape of the population distribution and the sampling distribution of the mean is known as:

(a) Central limit theorem (b) Central Place Theorem

(c) Level of Significance (d) Coefficient of Determination

76. 'Trade-off' principally means:

(a) A balance between two desirable but incompatible features

(b) A balance between multiple features

(c) Profitable option between two choices

(d) Profitable option between many choices

77. Statutory towns as per Census 2011:

(a) 3799 (b) 4041

(c) 3894 (d) 4382

78. Slums are reflection of:

(a) Improper use of land (b) Industrialization

(c) Illiteracy (d) All of the above

79. What is a mixed economy?

(a) Modern and traditional industries (b) Public and Private Sectors

(c) Foreign and domestic investments (d) Commercial and subsistence farming

80. What is Gross National Product?

(a) The total value of goods and services produced in the country

(b) The total value of all the transactions in the country

(c) The depreciation in the total value of goods and services produced in the country

(d) The total value of goods and services produced in the country and the net factor income from abroad

81. Which of the following activities comes under the tertiary sector?

(a) Agriculture, dairying, forestry

(b) Manufacturing of automobiles, construction of houses

(c) Communication and banking

(d) None of the above

82. As per Census 2011, there are _____ out growths in India

(a) 918 (b) 189

(c) 981 (d) 891

83. The primary driver of rapid urban growth, especially in developing countries, is:

(a) Suburbanization (b) Rural-to-urban migration

(c) Urban renewal (d) Declining rural areas

84. What is a primary focus of Urban Economics?

(a) Global trade patterns

(b) The spatial organization of firms and households within cities

(c) Monetary policy

(d) Agricultural subsidies

85. Which factor is generally considered crucial for locating industries?

(a) Proximity to leisure facilities

(b) Easy access to raw materials, labour, and transportation

(c) High population density

(d) Government tax incentives only

86. Under which policy do public sector enterprises set prices such that total revenue equals total cost (no profit, no loss)?

(a) No-Profit No-Loss Policy (b) Marginal Cost Pricing

(c) Profit-price policy (d) Price skimming

87. For a project activity of 7 days optimistic time, 8 days most likely time and 15 days pessimistic time, the expected duration is:

(a) 5 days (b) 13 days

(c) 9 days (d) 11 days

88. For public projects where the government fails to provide services all by itself,_____ is a form of project financing, wherein a private entity receives a concession from the government bodies to finance, design, construct, own and operate the service facility.

(a) Tax free construction
(b) Crowd Sourcing of funds
(c) Build - Operate- Transfer (BOT)
(d) Infrastructure Project Development Fund

89. One of the following method falls under non-deterministic approach or the probabilistic approach:

(a) PERT
(b) CPM
(c) GANTT chart
(d) Bar chart

90. The method of economic analysis to compare long run economic effects of a number of projects is called:

(a) Cost effectiveness analysis
(b) Cost-beneût analysis
(c) Fiscal impact analysis
(d) PPBS (Planning Programming Budgeting System)

91. The particular task performance in CPM is known as:

(a) Dummy
(b) Event
(c) Activity
(d) Contract

92. IRR (Internal Rate of Return) is a crucial financial metric representing the _____

(a) Annualized rate of return an investment is expected to generate
(b) Quarterly rate of return
(c) Break-Even point
(d) Annual interest

93. The 74th Constitutional Amendment Act makes provisions for_____

1. Constitution and Composition of a Municipality
2. Personnel system in a Municipality
3. Relations between elected members (executive) of a municipality and the bureaucracy

Choose the correct option from the options given below:

(a) Only 1
(b) 1 and 2
(c) 1 and 3
(d) 1, 2 and 3

94. What is the role of 'The Real Estate(Regulation and Development) Act?

(a) Slum improvement and up-gradation
(b) Urban poverty alleviation
(c) Facilitating land supply and market
(d) Provision of urban amenities and facilities

95. Which of the following statements regarding land acquisition as per the Act of 2013 is FALSE?

(a) Acquisition process is undertaken in consultation with Local Self Government
(b) There is provision for rehabilitation for all the affected families
(c) The purposes for which land can be acquired has been mentioned
(d) Central Government directly acquires the land and transfers it to concerned agency

96. The Right to Fair Compensation and Transparency in Land Acquisition , Rehabilitation and Resettlement Act 2013 covers which of the following issues:

1. Fair compensation
2. Transparency
3. Resettlement and Rehabilitation
4. Public health

(a) 1 and 4 (b) 1, 3 and 4

(c) 2, 3 and 4 (d) 1, 2 and 3

97. The Town Planning Committee comes under which among the following parts of Constitution of India?

(a) Part VII (b) Part IX

(c) Part lXA (d) None of them

98. Which of the following is not a part of the Planning Legislation introduced in India before independence?

(a) Sanitary Commission 1864 (b) The Bombay Improvement Trust 1898

(c) The Model Town Planning Act 1900 (d) The Bombay Town Planning Act 1915

99. Chapter VII of the Mizoram Urban and Regional Development Act,2025 talks about:

(a) Town Planning Scheme (b) Local Area Plan

(c) Development Plan (d) Grievance Redressal

100. The Institute of Town Planners, India (ITPI) was established in:

(a) 1950 (b) 1951

(c) 1948 (d) 1949

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